

MANAS AGARWAL

MERN-Stack Developer & Competitive Programmer

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Education

Bachelor of Technology in Computer Science and Information Technology

Jun 2022 – Jun 2026

Dr. A.P.J. Abdul Kalam Technical University

Work Experience

SDE Intern - BLG Technologies (P) Ltd.

Sep 2025 – Present

- Engineered responsive and scalable **frontend modules** for the Human Resources Management System (HRMS) platform using **React.js** and **Tailwind CSS**, improving UI load time by **25%**
- Integrated 30+ backend APIs** to enable real-time data flow and seamless interaction between frontend and server
- Collaborated with a **10-member team** to **develop 15+ reusable and scalable UI components** for key HRMS features like employee management, attendance, and payroll

Backend Developer Intern - Educase India

Mar 2025 – Jun 2025

- Implemented features in the **School ERP Software** to manage franchise operations, multi-school data, fee tracking, staff and student records
- Developed and tested **RESTful APIs in Node.js**, debugging and reviewing code to fix **25+ critical bugs** and **70+ minor bugs**, enhancing backend stability and performance
- Optimized **SQL queries** and **database indexing**, reducing query execution time by **20-30%** and improving API performance

Position of Responsibility

Programming Club, AKGEC

College Society

Senior Coordinator

Feb 2024 – Present

Led and collaborated with a **40-member team** to organize and deliver Data Structures and Algorithms workshops in C++, mentoring over 150 junior students while enhancing **presentation and public speaking skills**

Technical Projects

Chess Master | *WebRTC, Socket.io, Sequelize ORM, PostgreSQL* | 🌐 [Github](#) | 📺 [Video Demo](#) | 🔗 [Site Link](#)

- Built a real-time **WebRTC signaling server** in **Node.js** and **React.js** to support live voice chat enabling players to share strategies for mastering chess, serving 100+ players
- Designed a **match history module** using **PostgreSQL** to store move histories and **React.js** components for interactive analysis, enabling players to review and annotate past matches
- Implemented an **Elo-inspired rating algorithm** to adjust player ratings based on match outcomes and opponent ratings, ensuring accurate reflection of player skill
- Integrated OAuth 2.0 with **Passport.js** for secure login and used **Socket.io** for real-time connections between players and friends

VStream | *React.js, Redux Toolkit, Node.js, Express, MongoDB* | 🌐 [Frontend](#) | 🌐 [Backend](#) | 🔗 [Site Link](#)

- Designed and developed a **unified video and music streaming platform** using **MERN stack**, offering seamless media consumption in a single interface
- Optimized backend performance with **MongoDB Aggregation Pipelines**, achieving 30–50% faster data processing, and engineered efficient **pagination** for 70% faster fetch times
- Configured **JWT authentication** with **OTP-based email verification** for user registration
- Built a responsive and modular frontend using **React.js**, **Redux Toolkit** for state management, and **Tailwind CSS** for modern UI

Awards & Achievements

- Knight** (Rating : 2015) at LeetCode - [manas_agarwal16](#) | **Specialist** (Rating : 1566) at Codeforces - [manas_agarwal16](#)
- Solved **500+** problems on **Data Structures and Algorithms** and **600+** problems on **Competitive Programming**
- Achieved **2nd rank** among 350+ participants in the DSA contest and **3rd rank** among 300+ participants in the Competitive Programming contest at Ajay Kumar Garg Engineering College

Technical Skills

Programming Languages & Databases: C, C++, JavaScript, SQL, TypeScript, MongoDB, MySQL, PostgreSQL

Technologies & Frameworks: React.js, Redux Toolkit, Node.js, Socket.io, WebRTC, Sequelize ORM, Prisma ORM, Git